

1.Perform all Arithmetic Operations

```
a=10
b=20
print('addition = ', a+b)
print(f'subtraction = ',a-b)
```

2.Print integer in Positive Format

```
num=20
if(num<0):
    num=num*-1
print(num)
```

3.Check 2 integers are equal or not

```
num1=20
num2=30
if(num1==num2):
    print('numbers are equal')
else:
    print('numbers are not equal')
```

4.check user entered integer is Digit(1 number) or Number(2numbers)

```
n=20
if(n>=-9 & n<=9):
    print(n,'is a digit')
else:
    print(n,'is two digit')
```

5.take input from user and check its 2 digit or single digit number

```
n=int(input("enter a number : "))
if(n>=10 and n<=99 or n<=-10 and n>=-99):
    print(f'{n} is two digit')
else:
```

```
print(f'{n} is not two digit')
```

6.Check user entered number is multiple of 3 &5 or not

```
n=int(input('enter a number : '))
if(n%3==0 and n%5==0):
    print('number is divisible by 3 and 5')
else:
    print('number is not divisible by both 3 and 5 ')
```

7.Read month number from user & print it is valid or not

```
m=int(input('enter a month number : '))
if(m>=1 and m<=12):
    print('is a valid month number')
else:
    print('is not a valid month number ')
```

8.Read month number from user & print Corresponding How many Days in that month

```
month = int(input('enter a month number : '))

if(month == 1 or month == 3 or month == 5 or month == 7 or month == 8 or month == 10 or month == 12):
    print("Number of days is 31")
elif(month==4 or month==6 or month==9 or month==11):
    print("Number of days is 30")
elif(month == 2):
    print(" of days is 28")
else:
    print("Number of days is 30")
```

9.print results like Distinction ,1st class,2nd class,pass or Fail

```
print("Enter Subjects marks: ")
p = int(input('p : '))
c = int(input('c : '))
```

```
m = int(input('m : '))
b = int(input('b : '))
per = ((p + c + m + b) / 4)

if(p<35 or c<35 or m<35 or b<35):
    print(" Student is Fail")
else:
    if(per >= 85):
        print("Distinction")
    elif(per >= 60):
        print("First Class")
    elif(per >= 50):
        print("Second Class")
    else:
        print("Pass")
```

10. Check number is even or not

```
a=24
if(a%2==0):
    print('even')
else:
    print('odd')
```

11. Basics of loop print from 1 to n

```
n=10
i=1
while(i<=n):
    print(i)
    i=i+1
```

```
for i in range(5):
    print(i,end=" ")
```

12. print from n to 1

```
n=10
for i in range(n,0,-1):
```

```
print (i , end=" ")
```

```
i =10
while (i >= 1):
    print(i, end=' ')
    i = i - 1
```

13.multiplication table

```
num=12
for i in range(1, 11):
    print(num, 'x', i, '=', num*i)
```

14.print even numbers

```
for i in range(1,20):
    if i%2==0:
        print(i,end=" ")
```

15.sum of even

```
result = 0
for x in range(1, 10):
    if (x % 2 == 0):
        result = result + x

print("Sum of even number is ", result)
```

16.count number of digits in a number

```
n=int(input("Enter number:"))
count=0
while(n>0):
    count=count+1
    n=n//10
print("The number of digits in the number are:",count)
```

17.add 2 numbers using a function

```
def addNumbers(x,y):  
    sum = x + y  
    return sum  
  
output = addNumbers(12,9)  
print(output)
```

18.swap using funtion

```
def SwapTwoNumbers(a,b):  
    print("Before Swap: ",a,b)  
    a = a+b  
    b = a-b  
    a = a-b  
    return a,b
```

```
a,b = SwapTwoNumbers(17,24)  
print("After Swap: ",a,b)
```

```
//or    int temp = a;  
        a = b;  
        b = temp;
```

19.Read amount from user & print it in the Indian Currency Dimenssion

```
amt=int(input("Enter the numerical amount : "))  
print("2000rs --> ", int(amt/2000))  
amt=amt%2000  
print("500rs  --> ", int(amt/500) )  
amt=amt%500  
print("200rs  --> ", int(amt/200) )  
amt=amt%200  
print("100rs  --> ", int(amt/100) )  
amt=amt%100  
print("50rs   --> ", int(amt/50)  )  
amt=amt%50  
print("20rs   --> ", int(amt/20)  )  
amt=amt%20  
print("10rs   --> ", int(amt/10)  )
```

```
amt=amt%10
print("5rs    --> ", int(amt/5)    )
amt=amt%5
print("2rs    --> ", int(amt/2)    )
amt=amt%2
print("1rs    --> ", int(amt/1)    )
amt=amt%1
```

20.factorial

```
def factorial(n):
    fact = 1
    while(n!=0):
        fact = fact*n
        n = n-1
    print("The factorial is",fact)

inputNumber = int(input("Enter the number: "))
factorial(inputNumber)
```

21.Function to get sum of digits

```
def getSum(n):

    sum = 0
    while (n != 0):
        sum = sum + (n % 10)
        n = n//10
    return sum

n = 12345
print(getSum(n))
```

22.check number is pallindrome or not

```
n=int(input("Enter number:"))
temp=n
rev=0
while(n!=0):
    dig=n%10
    rev=rev*10+dig
```



```
    n=n//10
if(temp==rev):
    print("The number is a palindrome!")
else:
    print("The number isn't a palindrome!")
```

23.count even and odd numbers present in the list

```
list1 = [10, 21, 4, 45, 66, 93, 1]

even_count, odd_count = 0, 0

for num in list1:
    if num % 2 == 0:
        even_count += 1
    else:
        odd_count += 1

print("Even numbers in the list: ", even_count)
print("Odd numbers in the list: ", odd_count)
```

24.Python Program to check whether the given input is alphabet, number or special character

```
ch = input("Please Enter Any Character : ")

if((ch >= 'a' and ch <= 'z') or (ch >= 'A' and ch <= 'Z')):
    print("The Given Character ", ch, "is an Alphabet")
elif(ch >= '0' and ch <= '9'):
    print("The Given Character ", ch, "is a Digit")
else:
    print("The Given Character ", ch, "is a Special Character")
```

25.check str is vowel or consonant

```
l = input("Enter the character: ")
if l.lower() in ('a', 'e', 'i', 'o', 'u'):
    print("Vowel")
elif l.upper() in ('A', 'E', 'I', 'O', 'U'):
```

```
    print("Vowel")
else:
    print("Consonant")
```

26. count howmany capital ,lower,vowels,consonant,numbers in a string

```
name=str(input("enter the string"))
count=0
for i in name:
    if i.isupper():
        count=count+1
print("The number of capital letters found in the string is:-",count)
```

27. Find out How many 1 and 0 in a Given Number

```
gvn_num = 111000101
ones_cnt = 0
zeros_cnt = 0

while gvn_num:
    remindr = gvn_num % 10      # Calculate the value of the given number modulus 10 (which gives the last digit of the given Number)
    gvn_num = int(gvn_num/10)   # Calculate the value of the given number divided by 10 (which removes the last digit of the given Number)

    if remindr == 1:
        ones_cnt = ones_cnt+1

    if remindr == 0:
        zeros_cnt = zeros_cnt+1

print("The Count of 0's in a given number = ", zeros_cnt)
print("The Count of 1's in a given number = ", ones_cnt)

output:
The Count of 0's in a given number = 4
The Count of 1's in a given number = 5
```

28. Find the Sum of Natural Numbers


```
* n=5
sum=0
for i in range(1,n+1):
    sum=sum+i
print(sum)

* while(n!= 0):
    sum = sum+n
    n = n-1
print("Sum of natural numbers", "=", sum)

* sum = (number * (number+1))//2 #Using mathematical formula
```

29.print multiplication tables upto n

```
n=5
for j in range(1,n+1):
    for i in range(1,11):
        print(j , 'x', i , '=', (j*i))
    print()
```

30.print multiplication table of a given number

```
n=5
for i in range(1, 11):
    print(n, 'x', i, '=', (n * i))

number = 8
m = 1
while(m <= 10):
    print(number, "*", m, "=", number*m)
    m = m+1
```

31.Check Prime Number

A prime number is a positive integer greater than 1 that has no other variables except 1 and the number itself. Since they have no other variables, the numbers 2, 3, 5, 7, and so on are prime numbers.

```
number = 5
temp = False
# We will check if the number is greater than 1 or not
# since prime numbers starts from 2
if number > 1:

    for i in range(2, number//2):    # checking the divisors of the number
        if (number % i) == 0:
            temp = True            # if any divisor is found then set temp to true since it is not prime number
            break                  # Break the loop if it is not prime
if(temp):
    print("The given number", number, "is not prime number")
else:
    print("The given number", number, "is prime number")
```

32.Find the Factors of a Number and count number of factors

```
number = 10
count=0
print("printing the factors of given number : ")
for i in range(1,number+1):
    if(number%i==0):
        print(i)                #1 2 5 10
        count=count+1
print("Total Number of divisors are = ",count) #4
```

33.Finding Factorial of a Number

```
num = 7
res = 1

for i in range(1, num+1):
    res = res*i
    print(i,'! =',res)

print("Factorial of", num, "=", res)    # Factorial of 7 = 5040
```

output:

```
1 ! = 1
```

```
2 ! = 2
3 ! = 6
4 ! = 24
5 ! = 120
6 ! = 720
7 ! = 5040
```

34.Count the Number of Digits Present In a Number

```
* given_number = 27482
  digits_count = 0
  while (given_number > 0):
      given_number = given_number//10
      digits_count = digits_count + 1
  print("The total digits present in the given number=", digits_count)

* given_number = 27482
  print("The total digits present in the given number=", len(str(given_number)))
```

35.Reverse the given Number

```
given_num = 12345
reverse_number = 0
while (given_num > 0):
    remainder = given_num % 10      # getting the last digit
    reverse_number = (reverse_number * 10) + remainder
    given_num = given_num // 10

print("The reversed number =", reverse_number)
```

36.Find the Greatest Digit in a Number.

```
gvn_num = 154
str_numbr = str(gvn_num)
lst = list(str_numbr)
maxim_digit = max(lst)
print("The maximum digit in given number {" , gvn_num, "} = " , maxim_digit)
```

37. Print all Numbers in a Range Divisible by a Given Number

```
lower_limit = 20
upper_limit = 60
numb = 4
print("The numbers which are divisible by", numb, "from", lower_limit, "to", upper_limit, "are:")
for val in range(lower_limit, upper_limit+numb):
    if(val % numb == 0):
        print(val, end=" ")
```

output: The numbers which are divisible by 4 from 20 to 60 are: 20 24 28 32 36 40 44 48 52 56 60

38. Find Those Numbers which are Divisible by 7 and Multiple of 5 in a Given Range of Numbers

```
lowerlimit = 10
upperlimit = 200
print("The number which is divisible by 7 and multiple of 5 :")
for numb in range(lowerlimit, upperlimit+1):
    if(numb % 7 == 0 and numb % 5 == 0):
        print('Number =', numb)
```

output:

The number which is divisible by 7 and multiple of 5 :

```
Number = 35
Number = 70
Number = 105
Number = 140
Number = 175
```

39. Print Odd Numbers Within a Given Range

```
lowLimitRange = 1
uppLimitRange = 30
print('Odd numbers from', lowLimitRange, 'to', uppLimitRange, 'are :')
for interval in range(lowLimitRange, uppLimitRange+1):
    if(interval % 2 != 0):
        print(interval, end=" ")
```

40. Accept Three Digits and Print all Possible Combinations from the Digits

```
firstDigit = 1
secondDigit = 9
thirdDigit = 2

digitsList = []

digitsList.append(firstDigit)
digitsList.append(secondDigit)
digitsList.append(thirdDigit)
# Using nested loops
for i in range(3):
    for j in range(3):
        for k in range(3):
            if(i != j & j != k & k != i):
                print(digitsList[i], digitsList[j], digitsList[k])
```

Output:

```
1 9 2
1 2 9
9 1 2
9 2 1
2 1 9
2 9 1
```

41. Find Even Digits Sum and Odd Digits Sum Divisible by 4 and 3 Respectively

```
numb = 123452
stringnum = str(numb)

digtslst = list(map(int, stringnum))      # Create a list of digits say "digtslst" using map(),list(),int functions.
evn_sum = 0
od_sum = 0

for itr in range(len(digtslst)):
    if(digtslst[itr] % 2 == 0):
        evn_sum += digtslst[itr]
```

```
    else:
        od_sum += digtslst[itr]

if(evn_sum % 4 == 0 and od_sum % 3 == 0):
    print("yes, the even digits sum and odd digits sum of a given number{", numb, "} is divisible by 4 and 3 respectively.")
else:
    print("No, the even digits sum and odd digits sum of a given number{", numb, "} is not divisible by 4 and 3 respectively.")
```

Output: yes, the even digits sum and odd digits sum of a given number{ 123452 } is divisible by 4 and 3 respectively.

42.Count the Number of Odd and Even Digits of a Number at Even and Odd places

```
numb = 1787725620
stringnum = str(numb)

digtslst = list(map(int, stringnum))

evn_sum = 0
od_sum = 0

for itr in range(len(digtslst)):
    if(itr % 2 == 0):
        evn_sum += digtslst[itr]
    else:
        od_sum += digtslst[itr]

print("The sum of all digits at even places in a given number{", numb, "} =", evn_sum)
print("The sum of all digits at odd places in a given number {", numb, "} =", od_sum)
```

Output:

The sum of all digits at even places in a given number{ 1787725620 } = 23
The sum of all digits at odd places in a given number { 1787725620 } = 22

43.Count the Number of Odd and Even Digits

```
numb = 1237891
stringnum = str(numb)

digtslst = list(map(int, stringnum))
```



```
evn_count = 0
od_count = 0

for itr in range(len(digitslst)):
    if(digitslst[itr] % 2 == 0):
        evn_count += 1
    else:
        od_count += 1

print("The count of even digits in a given number{", numb, "} =", evn_count)
print("The count of odd digits in a given number{", numb, "} =", od_count)
```

Output:

The count of even digits in a given number{ 1237891 } = 2
The count of odd digits in a given number{ 1237891 } = 5

44. Print Palindrome Numbers in a Range in Python

```
gvn_lowrmt = 50
gvn_upprmt = 130
print("The palindrome numbers between", gvn_lowrmt, "and", gvn_upprmt, "are:")

for m in range(gvn_lowrmt, gvn_upprmt+1):
    given_num = m
    duplicate_num = given_num

    reverse_number = 0

    while (given_num > 0):
        remainder = given_num % 10
        reverse_number = (reverse_number * 10) + remainder
        given_num = given_num // 10

    if(duplicate_num == reverse_number):
        print(duplicate_num, end=" ")
```

Output: The palindrome numbers between 50 and 130 are: 55 66 77 88 99 101 111 121

45. Program to Find the Sum of Series $9+99+999+\dots+N$

```
gvn_num = 6
resultsum = 0

k = 9
for itr in range(1, gvn_num+1):
    resultsum += k
    k = (k*10)+9

print("The total sum of the series till the given number {", gvn_num, "} = ", resultsum)
```

Output: The total sum of the series till the given number { 6 } = 1111104

46. Print Series 1, 2, 4, 8, 16, 32...n

```
gvn_num = int(input("Enter some Random Number = "))
itr = 1

print("The above series till the given number{", gvn_num, "} is :")
while itr <= gvn_num:
    print(itr, end=" ")
    itr *= 2
```

Output:

Enter some Random Number = 50

The above series till the given number{ 50 } is :1 2 4 8 16 32

47. Print Series 1, 22, 333, 4444...n

```
gvn_num = 5
print("The above series till the given number{", gvn_num, "} is :")

for itr in range(gvn_num+1):
    for m in range(itr):
        print(itr, end="")
```



```
print(end=" ")
```

Output: The above series till the given number{ 5 } is : 1 22 333 4444 55555

48. Disarium Number

```
num = 135
ans = 0

digits = len(str(num))
dup_number = num
while (dup_number != 0):
    remainder = dup_number % 10
    ans = ans + remainder**digits
    digits = digits - 1
    dup_number = dup_number//10

if(num == ans):
    print(num, "is disarium number")
else:
    print(num, "is not disarium number")
```

Output: 135 is disarium number

49. Armstrong Number in Python

```
num = 153
ans = 0
digits = len(str(num))
dup_number = num
while (dup_number != 0):
    remainder = dup_number % 10
    ans = ans + remainder**digits
    dup_number = dup_number//10

if(num == ans):
    print(num, "is Armstrong number")
else:
```

```
print(num, "is not Armstrong number")
```

output: 153 is Armstrong number

Explanation: Here $1^3 + 5^3 + 3^3 = 153$ so it is Armstrong Number

50. Check Whether the given Number is Perfect Number or Not

appropriate divisors of 6 are 1, 2, 3.

Because the sum of these divisors equals 6 ($1+2+3=6$), 6 is considered a perfect number.

When we consider another number, such as 12, the proper divisors of 12 are 1, 2, 3, 4, and 6.

Now, because the sum of these divisors does not equal 12, 12 is not a perfect number.

```
def checkPerfectNumb(givenNumb):
    totalSum = 1

    for i in range(2, givenNumb//2 + 1):
        if givenNumb % i == 0:
            totalSum += i

    if(totalSum == givenNumb):
        return True

    return False

given_numb = 6
if(checkPerfectNumb(given_numb)):
    print("The given number", given_numb, "is perfect number")
else:
    print("The given number", given_numb, "is not a perfect number")
```

Generate Perfect Numbers in an Interval

```
def checkPerfectNumbr(givenNumb):
    totalSum = 1

    for i in range(2, givenNumb):
        if givenNumb % i == 0:
```

```
        totalSum += i

    if(totalSum == givenNumb):
        return True

    return False

lowlimrange = 1
upplimrange = 1000
print('The Perfect numbers in the given range', lowlimrange, 'and', upplimrange, 'are:')

for itrvalue in range(lowlimrange, upplimrange+1):
    if(checkPerfectNumbr(itrvalue)):
        print(itrvalue, end=' ')
```

Output: The Perfect numbers in the given range 1 and 1000 are: 1 6 28 496

51. Check Whether the given Number is Strong Number or Not

```
def checkStrongNumb(givenNumb):
    totalSum = 0
    tempNum = givenNumb

    while(givenNumb):    # using while to extract digit by digit of the given number
        s = 1
        factNum = 1
        remainder = givenNumb % 10

        while(s <= remainder):    # calculating the factorial of the digit(extracted by remainder variable)
            factNum = factNum * s
            s = s + 1

        totalSum = totalSum + factNum
        givenNumb = givenNumb//10

    if(totalSum == tempNum):
        return True
```

```
        return False

given_numb = 145
if(checkStrongNumb(given_numb)):
    print("The given number", given_numb, "is strong number")
else:
    print("The given number", given_numb, "is not a strong number")
```

Ex: 145 the sum of factorial of digits = 1 ! + 4 ! + 5 ! = 1 + 24 + 125

Check If A Number Is A Happy Number

Given Number = 320
Square of the digits = 3² + 2² + 0² = 13
Square of the digits = 1² + 3² = 10
Square of the digits = 1² + 0² = 1

```
def digitSquareSum(resltnumber):

    strnumbe = str(resltnumber)
    numbrlistdigits = list(map(int, strnumbe))

    sumsquaredigits = 0
    for digitvalu in numbrlistdigits:
        sumsquaredigits = sumsquaredigits+(digitvalu**2)

    return sumsquaredigits

numbr = 100
while(reslt != 1 and reslt != 4):
    reslt = digitSquareSum(reslt)

if(reslt == 1):
    print('The given number [' , numbr, ' ] is a happy number')
else:
    print('The given number [' , numbr, ' ] is not a happy number')
```

Print all Happy numbers within a given range

```
def digitSquareSum(resltnumber):
    strnumbe = str(resltnumber)
    numbrlistdigits = list(map(int, strnumbe))
    sumsquaredigits = 0

    for digitvalu in numbrlistdigits:
        sumsquaredigits = sumsquaredigits+(digitvalu**2)

    return sumsquaredigits

def checkhappypnumb(numb):
    rest = numb

    while(rest != 1 and rest != 4):
        rest = digitSquareSum(rest)

    if(rest == 1):
        return True
    else:
        return False

lowlimrange = 19
upplimrange = 145
print('The happy numbers in the given range', lowlimrange, 'and', upplimrange, 'are:')

for l in range(lowlimrange, upplimrange+1):
    if(checkhappypnumb(l)):
        print(l, end=' ')
```

Output:

The happy numbers in the given range 19 and 145 are:

19 23 28 31 32 44 49 68 70 79 82 86 91 94 97 100 103 109 129 130 133 139

Print all the Prime Numbers within a Given Range

```
givenNumbr = 95
print('The prime numbers from 2 to upper limit [' , givenNumbr, ' ] :')

for valu in range(2, givenNumbr+1):
    cntvar = 0
    for divi in range(2, valu//2+1):
        if(valu % divi == 0):
            cntvar = cntvar+1

    if(cntvar <= 0):
        print(valu, end=' ')
```

Output:

The prime numbers from 2 to upper limit [95] :
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89